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UNIVERSITI PENDIDIKAN SULTAN IDRIS

SULTAN IDRIS EDUCATION UNIVERSITY

UNIVERSITI PENDIDIKAN SULTAN IDRIS

FINAL EXAMINATION
SEMESTER 2 SESSION 2015/2016

CODE : MES3103

COURSE : SOFTWARE VERIFICATION AND VALIDATION

DATE : 15 JUN 2016

DURATION : 2 HOURS

INSTRUCTIONS

1. This question paper consists of **TWO (2)** sections only, namely **Section A** and **Section B**.
2. **Section A** : Answer **ALL** questions in the answer booklet provided.
3. **Section B** : Answer **ALL** questions in the answer booklet provided.
4. This question paper will be collected at the end of the examination.

This question paper consists of 5 printed pages including front page

PROGRAMME : _____

YEAR : _____ CLASS GROUP : _____

REGISTRATION NO : _____ | | | | | | |

IDENTITY CARD NO : | | | | | | | | | | | | |

LECTURER : DR. SHAMSUL ARRIEYA ARIFFIN

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

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SECTION A (60 MARKS)

Instructions: Answer **all** questions by writing your answer in the booklet provided.

1. Muhammad Hidayat is an exceptional programmer. Testing his software modules reveals very few errors, far fewer than the team's average. He keeps his schedule promptly, and only rarely is he late in completing his task. He always finds original ways to solve programming difficulties, and uses an original, individual version of the coding style. He dislikes preparing the required documentation, and rarely does he do it according to the team's templates. A day after completing a challenging task, on time, he was called to the office of the department's chief software engineer. Instead of being praised for his accomplishments (as he expected), he was warned by the company's chief software engineer that he would be fired unless he began to fully comply with team's coding and documentation instructions.

   Do you agree with the position taken by the department's chief software engineer? If yes, could you suggest why his/her position was so decisive?

   [5 marks]

2. There are **THREE (3)** major differences between software products and other industrial products.

   a. Identify and describe the **three (3)** differences.

   b. Discuss the ways in which these differences affect Software Quality Assurance (SQA).

   [5 marks]

3. a. Define software error, software fault and software failure. Explain the differences between these undesirable software statuses.

   [3 marks]

   b. Suggest a situation where a new type of software failure ("bug") appears in a software package that has been serving 300 clients for the first time six years since the software package was first sold to the public.

   [2 marks]

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4. National Software Providers is very interested in the newly developing area of BI(Business Intelligence) for electronic commerce firms. As the company is very keen to gain experience in this area, they were especially interested in winning a tender issued by one of the leading cosmetics manufacturers. The proposal team estimated that in order to win the contract, their proposal should not exceed the sum of $650,000.Accordingly, their quotation was $647,000. As all the team members were aware that the cost of completing the project by the company's inexperienced development department would substantially exceed this sum, they decided that there was little use in investing efforts to estimate the actual costs of the project.

   Do you agree with the team's decision not to estimate the project's costs? If you disagree, what are your arguments in favor of estimating the costs?

   [5 marks]

5. Some professionals claim that increased software usability necessarily involves decreased efficiency. Others claim no dependence between software efficiency an usability.

   a. Do you agree with the first or the second group? Discuss your answer.

   [5 marks]

   b. For each of the following task situations and interfaces, identify the likely guideline or guideline that has been violated. A user sees the following button choices in a print interaction:

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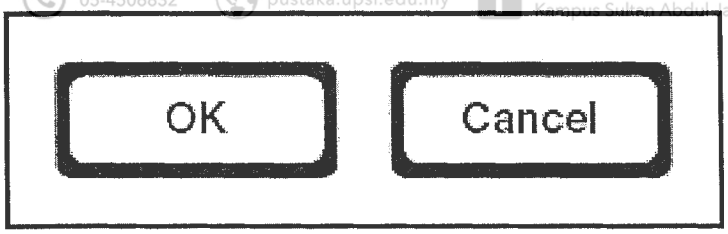

Figure 1

Then the user sees the following button choices in an interaction to insert clip art:

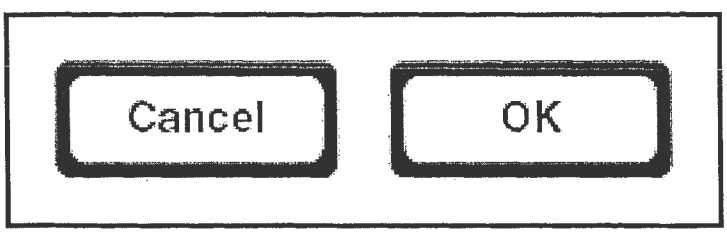

Figure 2 ***[5 marks]***

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6. a. List and briefly describe **four (4)** various causes of software errors.

[8 marks]

b. Classify the causes of error according to the groups responsible for the error: the client's staff, the systems analysts, the programmers, the testing staff – or is it a shared responsibility belonging to more than one group?

[7 marks]

7. a. What are the **three (3)** factor categories of McCall's factor model? Provide **two (2)** factors for each categories and the definition.

[10 marks]

b. Choose **one (1)** factor that best fits each requirements below according to McCall's eleven factors and to which factor categories does the factor belongs.

Requirements	**McCall's requirement factor**	**Factor category**
The probability a software system will be found in a state of failure during peak hours (9am-4pm) is required to be below 0.5%.		
The software package developed for the Linux operating system should be compatible for applications in a Windows NT environment.		

[5 marks]

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SECTION B (40 MARKS)

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Instructions: Answer **all** questions by writing your answer in the booklet provided.

1. Assume that the Gemilang Trade Company is going to implement a company-wide instant messaging system (**IMS**) as part of its technology plan for this year. Gemilang Trade has never had companywide instant messaging before and the company is considering several instant messaging systems.

   a. Develop a usability assessment strategy to analyze and design an evaluation for the candidate instant messaging systems (**IMS**). What are the usability attributes will you focus on? Will your tests be summative or formative? What type of assessment will you perform?

   [10 marks]

   b. At the low-level design questions for your evaluation, what types of measurements and usability assessments will you use? For each, be sure to identify *what* and *how* you are measuring. What procedure will you follow?

   [10 marks]

2. Define the following terms: *usability assessment, analytic evaluation, evaluation by experts, evaluation by users, user testing, benchmark task, objective data, subjective data, qualitative data, quantitative data, controlled setting, formative evaluation, summative evaluation, GOMS analysis, heuristic evaluation, cognitive walk-through, QUIS, clickstream data,* and *verbal protocols.*

   [20 marks]

END OF QUESTIONS

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